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1. Skim through the topics of the accepted papers for the Vis 2026 conference: https://ieeevis.org/year/2026/program/content/.

Which topic sounds most interesting to you, so you would like to explore their paper and learn more, or, if possible, attend their presentation? Why? Justify your answer.

As I reviewed the VIS 2026 program, some topics that particularly interested me included artificial intelligence, cyber topics, and uncertainty visualization. One paper that me was "Evaluating Human-LLM Alignment on Visualization Preference," as I was curious about ways an AI-produced model might select a visualization similar to the one a human user would select. Another area of interest to me, given my background in cybersecurity, was "CUBE-ism," as I was curious about visualizations that could be useful for making decisions about which cyber events to investigate further.

The paper I'd most like to read is "A General Approach to Visualizing Uncertainty in Statistical Graphics" by Petek, Nabergoj, and Štrumbelj. In this paper, the authors sample from the data distribution several times and produce full visualizations using different samples that can be combined into a single visualization to show the overall data distribution. The author released a python package, bootplot, which I am interested in using a feature is the inclusion of separate charts showing variation between samples. I am familiar with other techniques for visualizing uncertainty, including error bars, and wanted to learn more about visualizing uncertainty in pie charts and tables.

This project reminded me of my Quantum MNIST project. On my Quantum MNIST project, one way to communicate the results of different machine learning methods was to report accuracy statistics. A potential drawback of reporting accuracy statistics is that it's hard for your audience to understand how variable the accuracy statistic is likely to be in repeated experiments. If given the chance to hear about this project at a conference or presentation, I would ask the presenters how they suggest communicating results from hybrid projects, including information about variability. The authors of this project have created a Python package that I am interested in using in repeated experiments to see if it improves communication of variability. (https://github.com/davidnabergoj/bootplot)

2. Analyze 5 attached papers of the winners of the Vis 2025 conference. What technologies/libraries/frameworks did they use? What data did they use? You can use AI for a PDF summary or Q&As. Why do you think they won?

1. **NLI4VolVis**
    - *Technology and data*. The authors used editable 3D Gaussian Splatting to build 3D scenes whose individual parts could be changed in real time. They

used SAM 2 and other methods to identify those parts, CLIP to match them to a user's words, and LLM agents to carry out the requests. The chat supports GPT-4o, DeepSeek-V3, and Llama 3.2; data was prepared in ParaView, and it ran on an NVIDIA RTX 4090 at over 100 FPS. They tested the system on eight datasets, including scans of a backpack, a carp, and human organs, as well as simulations of a hurricane, mantle temperature, and a supernova.

- *Why I think it won*. The example that stood out to me was asking to see a snake inside an egg. The system made the egg partly transparent, so the snake was visible. I can see how asking for what you want in plain language would make 3D data easier to explore, especially if you're new to the software. In the study with eight participants, about 88% of interactions used natural language.

2. **CausalMap and sentiment contagion**
    - *Technology and data*. The authors built CausalMap. They used Latent Dirichlet Allocation (LDA) to group posts into topics, an LLM to label sentiment, and a Bayesian network fitted with Fast Greedy Equivalence Search to model how sentiment might spread between topics. They tested it on two datasets, one about the NBA and basketball and one about U.S. presidential discussion. Each had more than 200,000 posts.
    - *Why I think it won*. I liked that the map lets you follow a possible path of influence over time and then open the posts behind it. Putting time on the horizontal axis also made a measurable difference. Analysts finished tasks faster and used fewer clicks and other interactions than they did with the animated version.

3. **Beyond Problem Solving**
    - *Technology and data*. The researchers interviewed 11 experienced visualization designers over Zoom and used Dovetail to transcribe and organize their notes. They looked for patterns across interviews, project diaries, sketches, and prototypes. They also gave everyone the same open-ended design challenge using the London Stage Database.
    - *Why I think it won*. This one reminded me of making charts for class. Sometimes I start with a question, make a rough chart, and realize there's a better question in the data. I think the paper stood out because the researchers used sketches and diaries to show how that change happens

while people are still designing.

4. **ReVISit 2**
    - *Technology and data*. ReVISit 2 is an open-source framework for running online studies. It uses JSON to define study steps, supports React and Vega for interactive tasks, and offers Python tools for building studies. The authors recruited 460 participants across three studies on judging correlation, designing chart textures, and searching within visualizations. They collected task responses and records of how participants interacted with the charts. Study data is saved with Google Firebase or Supabase.
    - *Why I think it won*. The replay feature stood out to me. If a participant gives an unexpected answer, I'd want to see what they clicked and what the chart looked like at that moment. Sharing the study code and settings also makes it easier for another researcher to repeat the experiment.

5. **They Aren't Built for Me**
    - *Technology and data*. The authors used swell paper, which raises printed lines and shapes when heated, to create 80 tactile bar, pie, bubble, and stacked bar charts. Eleven blind or low-vision participants estimated ratios by touch. The researchers collected their answers, response times, and comments, and four participants joined a follow-up group interview.
    - *Why I think it won*. What stuck with me was how participants used their fingers like calipers to compare lengths. Comparing the area of two bubbles took a different approach. I think the paper stood out because it showed how people actually read charts by touch and used those observations to suggest better ways to design tactile graphics.

What stood out to me is that all five papers started with a real problem people face when creating, reading, or using visualizations. They backed up their ideas with evidence from experiments, interviews, or examples from real projects. Whether the authors were making 3D data easier to explore or studying how people read charts by touch, they paid attention to how people interact with visualizations. I think that's what made these papers stand out.

3. Attempt to reproduce some of the paper's visualizations from the papers mentioned (if the code is available on GitHub). Alternatively, <u>you can choose any other paper to reproduce their Viz from Vis 2025,</u> and if you cannot obtain the PDF, I can assist with that.

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https://vc.cs.uni-magdeburg.de/assets/publications/2026/Friesecke_2026_VIS.pdf

Uncertainty-Aware PCA Revisited *Lukas Friesecke, Christian Braune, Christian Rössl, and Holger Theisel*

It is on the VIS 2025 listing.

- Locate the authors' public code repository, if available, and provide its URL.

https://github.com/lfriesecke/uncertainty-aware-pca-revisited/

- Follow the setup instructions and attempt to run the visualization.

- Record dependencies, errors, changes, and results; include an original-versus-reproduced screenshot.

With the help if Antigravity, Here is our information errors, changes and results and comparison screenshot. https://cps5745wk9as2.vercel.app/

Dependencies:

Item	Authors (README/ Requirements.txt)	My Machine
Python	3.13.5	3.10.10
Numpy	2.3.1	2.2.6
Matplotlib	3.10.3	3.10.3
Contourpy, pillow, fonttools, kiwisolver, etc	Pinned in requirements.txt	Same versions

Errors and changes:

1. *NumPy would not install.* Running pip install -r requirements.txt failed with *"No matching distribution found for numpy==2.3.1"*. numpy 2.3 needs Python 3.11 or newer, and I have 3.10.
   Change: I made requirements_py310.txt with numpy pinned to 2.2.6. Everything else installed at the pinned versions, and the authors' code ran with no edits.
2. *Runtime was too long.* The paper uses U = 1,000,000 samples. On my laptop, U = 10,000 already took 75 s for IRIS, so the full run would take about 2 hours per dataset.
   Change: I used U = 100,000 and kept every other setting from their examples (bins = 250, R = 0.2, the same axis limits, scale matrices, colors, and random seed).
3. *No glyph renderer.* The glyph example only writes an .off 3D mesh file. The tool the authors used to render it is not in the repo.

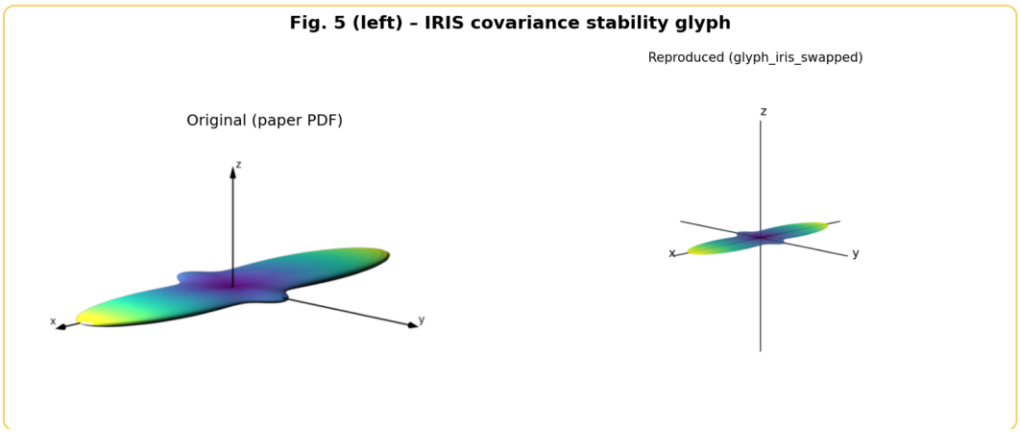
Change: I wrote reproduce_glyphs.py. It runs their Glyph class with 91 × 181 samples and draws the mesh in matplotlib using the viridis colormap.

4. *The glyphs came out rotated.* My first renders did not match the paper. In glyph.py there is a commented-out line, idx = np.array([2, 1, 0]) # TODO: ask if wanted, that reverses the eigenvector order. Without it, the first eigenvector lands on the x-axis instead of the z-axis the paper describes.
   Change: my script swaps the x and z coordinates of the mesh, which has the same effect. After that, both glyphs match the paper.
5. ***Mixed-up iris names (a bug in the authors' code).*** When I printed the species names that the code attaches to the Iris data, they came out as "Versicolor, Virginica, Virginica" instead of "Setosa, Versicolor, Virginica." The loader saves each group of flowers under the next species' name. The numbers are correct and in the right order, so the pictures are not affected; only the name labels are wrong.
   Change: I used the correct names on my interactive page and did not edit their code.
6. *Minor warning.* Matplotlib said its config folder was not writable. Setting MPLCONFIGDIR to a local folder fixed it.

Results:

Figure 5 – IRIS glyph

✨ Fig. 5 – IRIS covariance stability glyph

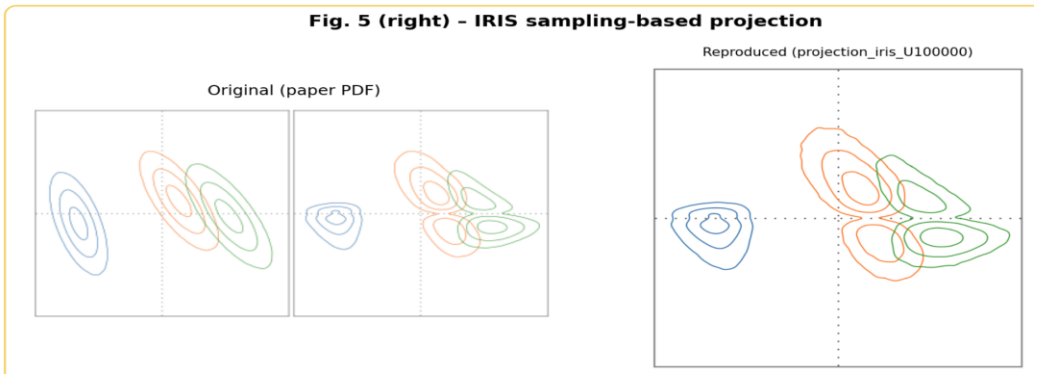


Runtime: <1 min

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Figure 5 – IRIS projection

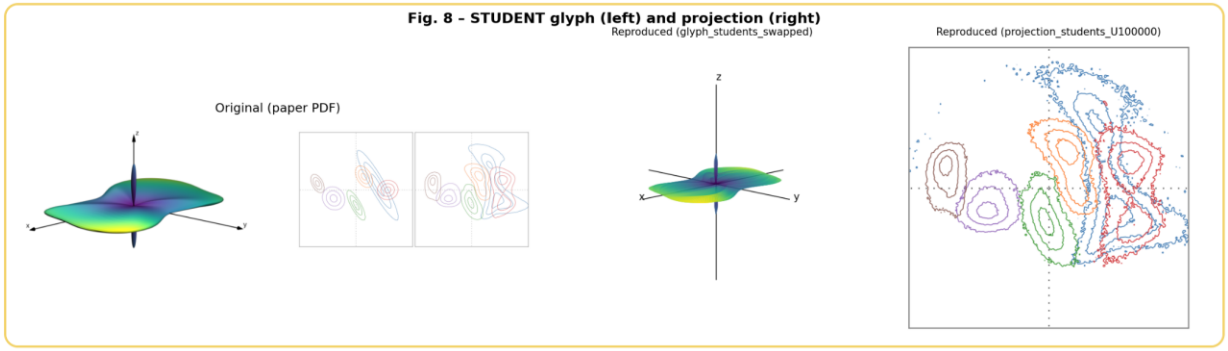
✨ Fig. 5 – IRIS sampling-based projection



Runtime: 10.8 minutes

Figure 8 – STUDENT glyph & STUDENT Projections

✨ Fig. 8 – STUDENT glyph and projection

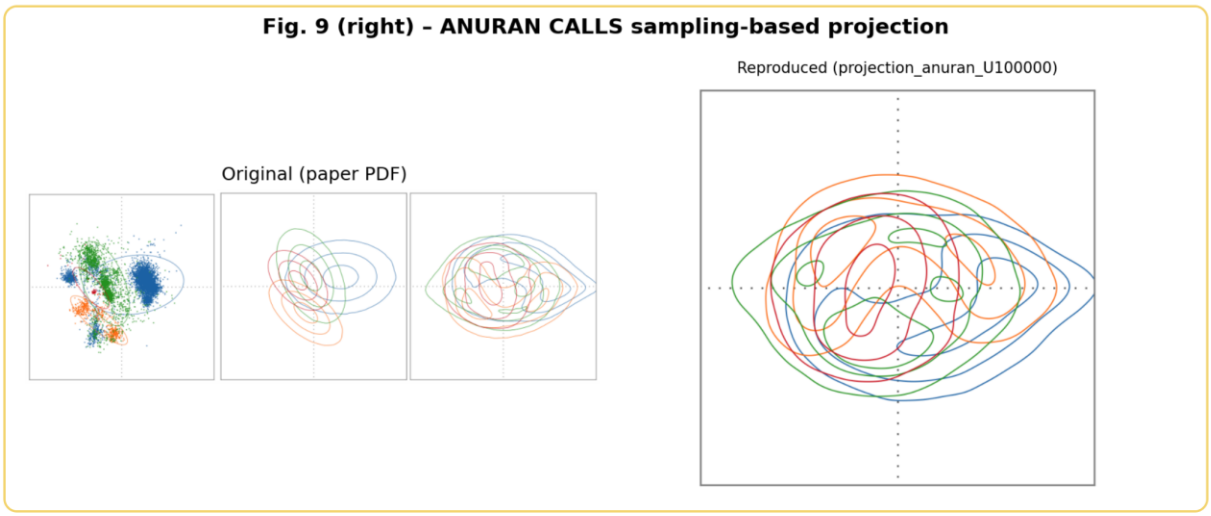


Runtime:<1 Min & 44.5 mins

Figure 9 right - ANURAN CALLS

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✨ Fig. 9 – ANURAN CALLS projection



Runtime: 5.2 mins

You can obtain extra points for a mini-presentation of reproduced visualizations.

Extra credit: mini-presentation (WK9AS2_presentation.pdf) and an interactive website where you can spin the 3D glyphs and watch the PCA picture change for each random version of the data: https://ponzek.github.io/CSP5745WK9AS2/